DATA ENGINEERING | BUSINESS INTELLIGENCE

Disaster Data Engineering & Intelligence Platform

Ten years of earthquakes and storms, from raw public data to a 4-page Power BI dashboard

1.7M
EARTHQUAKES

693K
STORM EVENTS

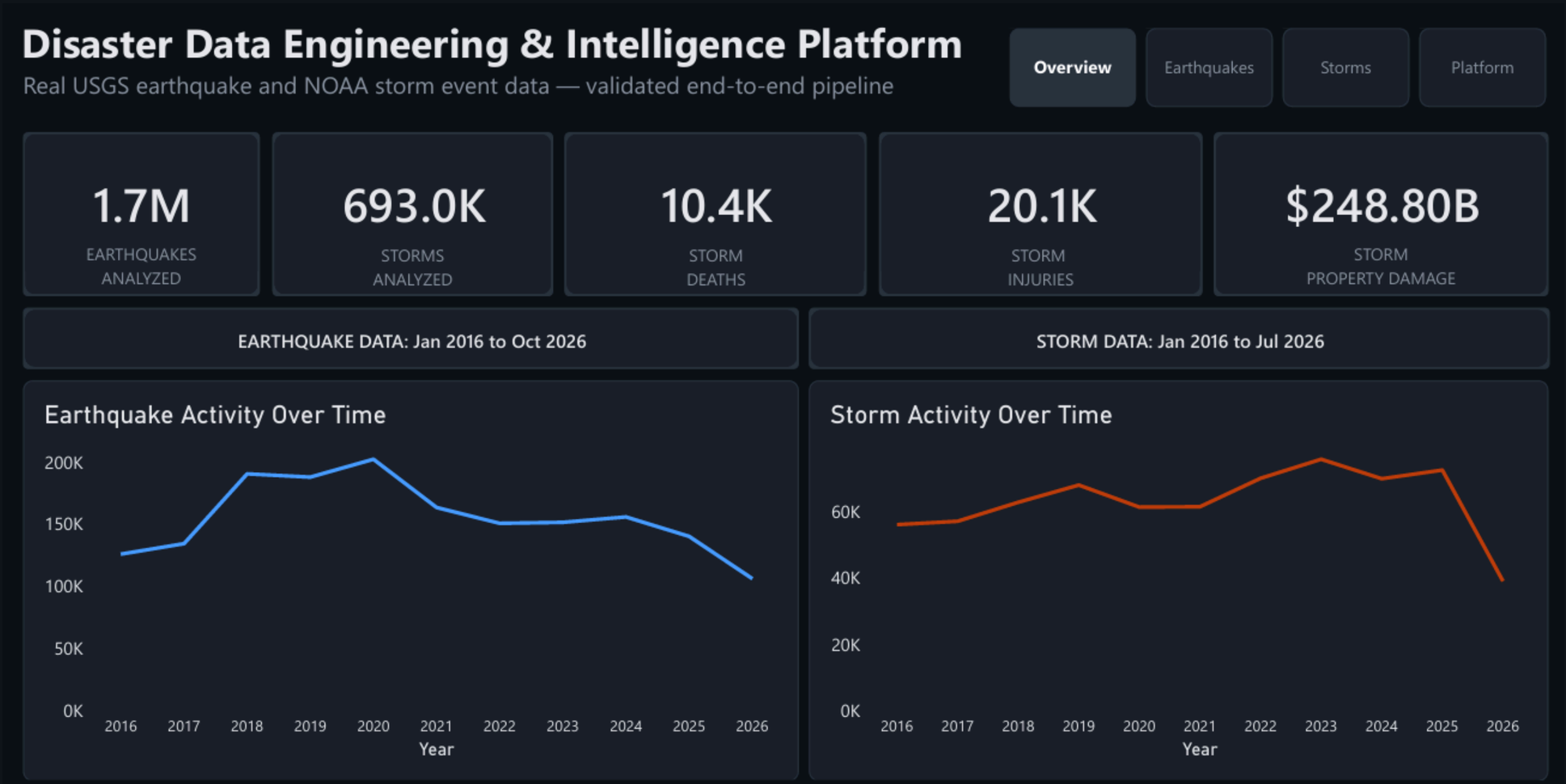
155+
AUTOMATED TESTS

Python • PostgreSQL • SQL • Power BI • DAX • pytest

DISASTER DATA ENGINEERING & INTELLIGENCE PLATFORM

One platform, two data sources

USGS earthquakes and NOAA storms, side by side from 2016 to 2026



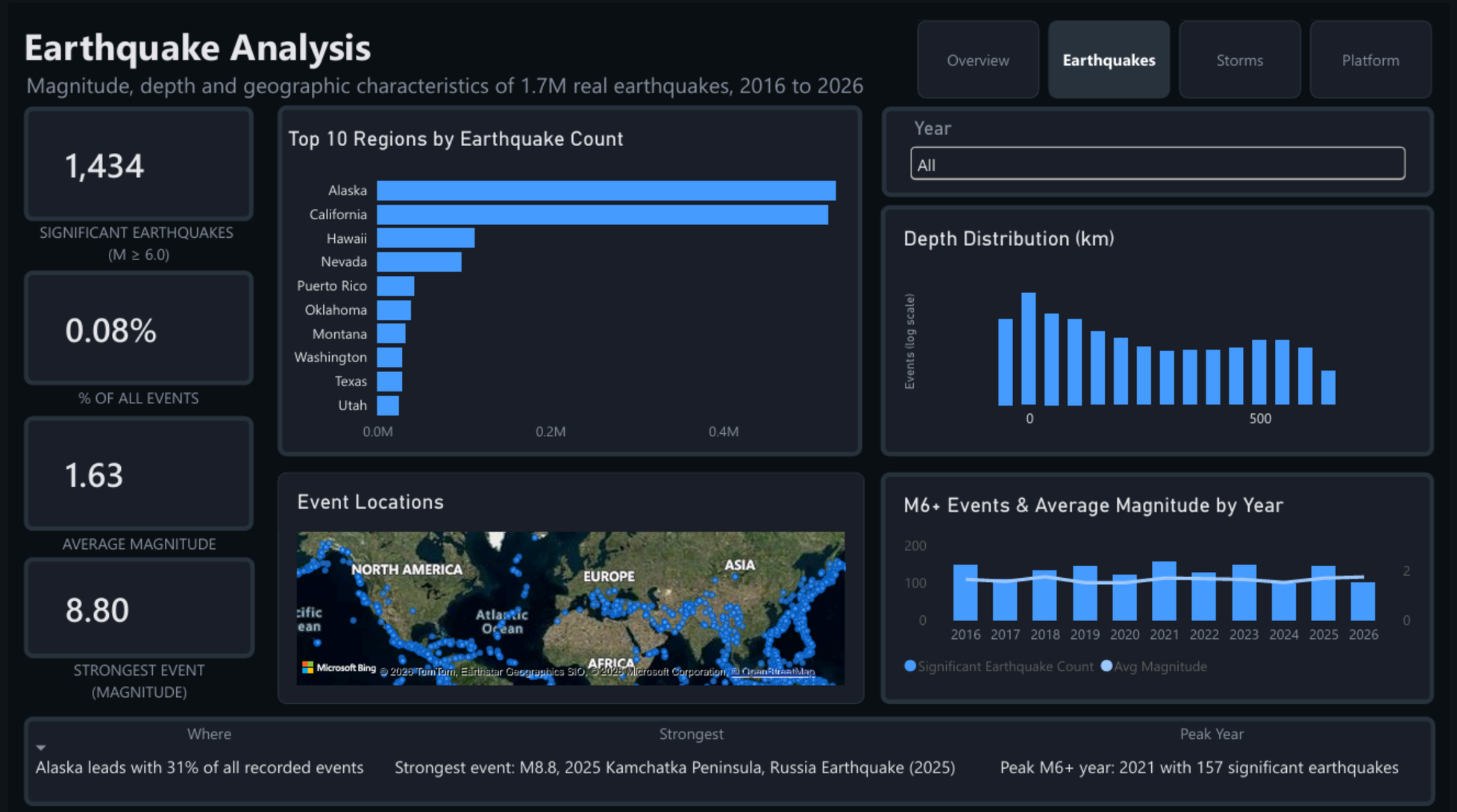
Data through 1 Oct 2026 (earthquakes) and 1 Jul 2026 (storms). The latest year is year-to-date and updates with each pipeline run.

Note: earthquakes and storm events are analyzed independently. No direct event-level correlation is drawn between the two sources, as no defensible geographic or temporal join key exists between USGS and NOAA records.

DISASTER DATA ENGINEERING & INTELLIGENCE PLATFORM

Alaska holds 31% of all recorded earthquakes

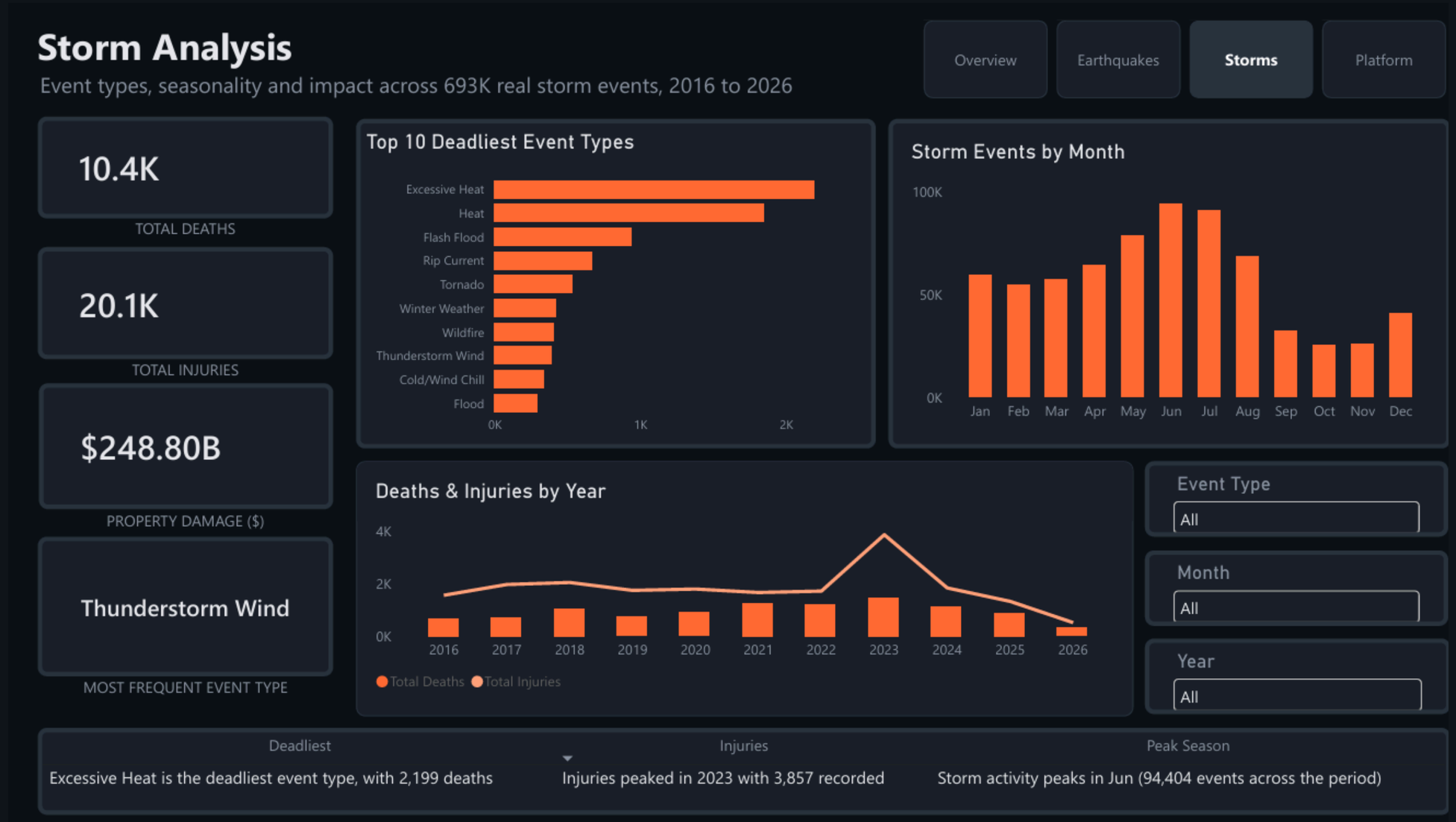
The strongest event was M8.8 near Kamchatka, Russia in 2025



DISASTER DATA ENGINEERING & INTELLIGENCE PLATFORM

Heat kills more people than tornadoes

Excessive Heat caused 2,199 deaths, the most of any storm type



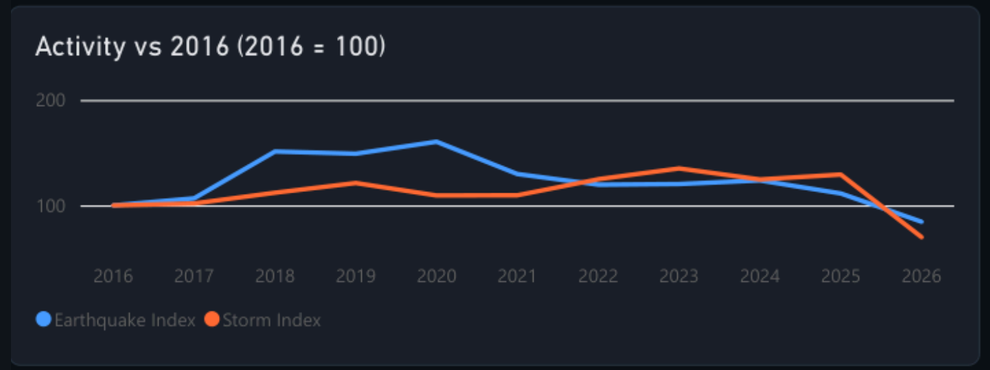
DISASTER DATA ENGINEERING & INTELLIGENCE PLATFORM

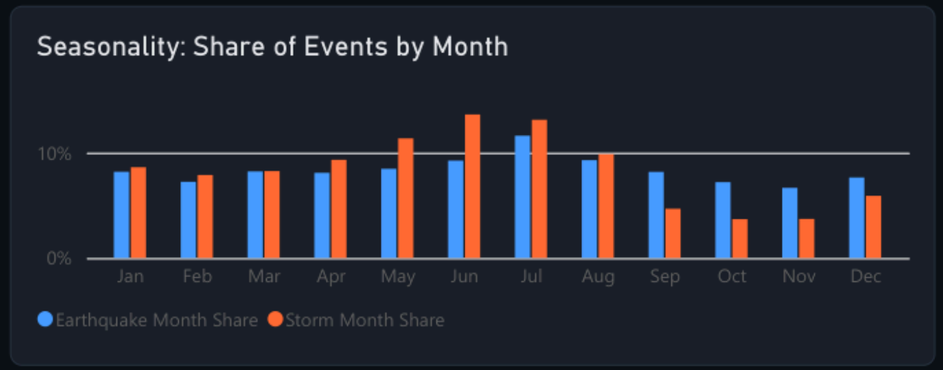
Earthquakes have no season. Storms do.

Earthquakes stay at 7-12% of events every month, while storms peak in June at 14%

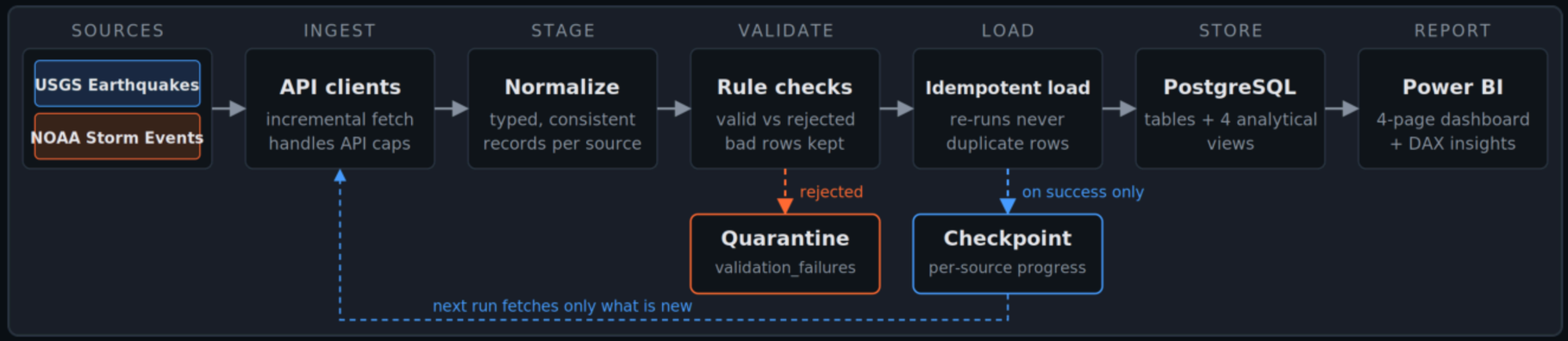
Platform Overview

Overview | Earthquakes | Storms | **Platform**





DATA PIPELINE



Earthquakes	Storms	Seasonality
Earthquake activity peaked in 2020 (+60% vs 2016)	Storm activity peaked in 2023 (+35% vs 2016)	Storms peak in Jun (14% of events); earthquakes stay between 7% and 12% every month

DATA PIPELINE

Built to fail safely

Incremental checkpoints, idempotent loads, a quarantine for bad records, and failure isolation between sources



CHECKPOINTS

Resume safely after successful progress

IDEMPOTENCY

Repeat runs without duplicating loaded data

QUARANTINE

Bad records are isolated instead of silently loaded

Python • PostgreSQL • SQL • Power BI • DAX • pytest

github.com/Sravani-2205/disaster-data-engineering-platform